Kick-off Diagnostic · L3-4 — Dividing Polynomials

Grade 10

8 questions · 30 seconds each · about 4 minutes · Diagnose phase (6 min of 60)

Pre-session anchor (stations 1 and 2 of the pre-session say this, so items 2 and 6 test it).
The one idea: Dividing $p(x)$ by $x-a$ leaves the remainder $p(a)$, and a remainder of 0 means $x-a$ is a factor; synthetic division finds the quotient and remainder using only the coefficients.
The mistake most students make: Writing $-a$ instead of a in the synthetic division box, or forgetting a zero for a missing power.
Readiness items 3, 4, 5, 7, 8 test: evaluating a polynomial at a number, factoring a quadratic, and reading coefficients.
Wayground set-up: timer 30 s on every question; shuffle options on, questions off; hide the leaderboard; show answers after the quiz, not after each item. Item 1 is a poll (no points). Items 2 to 8 are 1 point each. Tags below: **maroon** = tells you about the pre-session, **navy** = readiness for today. **Green** = correct.

1 POLL · 30 s — SELF-REPORT

Be honest: how much of the pre-session task did you do?

A. All of it, including the readiness check
B. I watched and read but skipped the check
C. I opened it but did not finish
D. I did not open it

Signal. Ungraded. Tally it first; it is the self-report half of the pre-session test.

2 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

In the pre-session, you divided by $x-2$ with synthetic division. Which number went in the box?

✓ **A.** 2
B. -2
C. 1
D. 0

Signal. Wrong → the pre-session misconception: use a, not $-a$. B = took the sign of the constant term; C = the leading coefficient; D = the remainder.

3 FILL IN THE BLANK · 30 s — READINESS

Let $p(x) = x^3 - 2x + 1$. Then $p(2) =$ _____

Accept: 5

Signal. $3 = 2^3$ replaced by $2 \cdot 3$ or an arithmetic slip; -3 = sign slip. Substituting is not yet secure.

4 CHECKBOX · 30 s — READINESS

For $p(x) = x^2 - 5x + 6$, select the **two** true statements.

✓ **A.** $p(2) = 0$
✓ **B.** $p(3) = 0$
C. $p(1) = 0$
D. $p(-2) = 0$

Signal. C = $1 - 5 + 6 = 2$; D = sign error on $-5x$. This is the factor idea that today's theorem names.

5 MATCH · 30 s — READINESS

Match each division with its quotient.

$(x^2 - 9) \div (x - 3)$ → $x + 3$
$(x^2 - x - 6) \div (x - 3)$ → $x + 2$ (shown shuffled in Wayground)
$(x^2 + x - 2) \div (x + 2)$ → $x - 1$

Signal. Mixing the first two = factored $x^2 - x - 6$ with the wrong pair; $x + 2$ for the third = did not check the factor pair.

6 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

The pre-session said a remainder of 0 tells you…

✓ **A. the divisor is a factor**
B. the polynomial has no roots
C. the quotient is 0
D. the divisor is 0

Signal. Wrong → skipped station 4's last prompt. C = confused remainder with quotient.

7 REORDER · 30 s **READINESS**

Put the steps in order to test whether $x - 1$ is a factor of $x^3 + 3x^2 - 4$.

Correct order: write the coefficients $1,\ 3,\ 0,\ -4$ (zero for the missing x) → bring down 1, then multiply by 1 and add: $1,\ 4,\ 4,\ 0$ → the last number, the remainder, is 0 → so $x - 1$ is a factor

Signal. Wrong → the placeholder zero is the gap; without it the coefficients shift.

8 DROPDOWN · 30 s **READINESS**

What is the remainder when $2x^2 + 3x - 5$ is divided by $x + 2$?

✓ **A.** -3 B. 9

C. 3 D. -5

Signal. B = used $p(2)$ for $x + 2$; C = sign slip in $p(-2)$; D = the constant term.

Read the results (90 seconds)

Readiness R **Items 3, 4, 5, 7, 8 (out of 5). 0–2** → re-teach substituting into a polynomial and the synthetic division steps with me. **3–4** → straight to the Remainder and Factor Theorems; fix the missed item in passing. **5** → start the Investigate prompt now.

Pre-session P **Items 2 and 6 (out of 2), read with the item 1 poll.**

What you see	Reading	What you do
Poll "all" or "read", $P = 2$	Prepared	Nothing extra. Use their R score to route.
Poll "all" or "read", $P = 0$	Claimed, did not retain	Two-minute recap of the one idea at the start. Ask them to restate it in their own words.
Poll "opened" or "did not open"	Task not done	No penalty. Pair with a prepared partner for Targeted Instruction; note it for your task tracker.
$R \leq 2$ and $P = 2$	Content gap	Re-teach group. The task was done, so the idea needs a different explanation.
$R \geq 4$ and $P \leq 1$	Task skipped, skills fine	Move on. Chase the task, not the content.

Class-level rule. If more than 30% of the class chose "opened" or "did not open", or the class average on P is below 1, spend three minutes recapping the one idea above before Targeted Instruction. If more than 30% score $R \leq 2$, the whole class takes the re-teach route first.